Pedestrian push buttons and signs shall be securely attached to the pole. Pedestrian pushbutton signs shall be temporarily covered until the signal is in operation.

635.17.03 Luminaires and Photocells - Luminaires and photocells shall be installed as per manufacturer's instructions or as noted on the Design. Luminaires shall be installed level. Where the luminaire has a multi-tap ballast, the Contractor shall verify the service voltage and adjust the luminaire voltage tap to suit.

When installing flat glass cobra head roadway luminaires the flat glass lens shall be oriented parallel to the roadway surface to reduce glare.

Sign luminaires will require different aiming depending on the manufacturer. Aiming angles shall be noted on the Design.

Photocell eyes shall be aimed north.

635.17.04 Streetname Signs - Streetname signs shall be bolted directly to the pole arms in accordance with Drawing SS SP635-3.2.1.

635.17.05 Audible Signals -Audible Signals shall be installed on pedestrian signal heads in accordance with Drawing SS SP635-2.3.8. The Contractor shall aim and adjust the audible signal heads to the satisfaction of the Designer and the Ministry Representative.

635.17.06 Emergency Vehicle Pre-emption Equipment – Emergency vehicle pre-emption equipment shall be installed in accordance with the Contract, Design and manufacturer's instructions. The Contractor shall undertake all commissioning and set-up of the equipment under the direction of the supplier technical representative.

635.17.07 Small Overhead Signs - Small overhead signs are classed as signs 1200 mm x 900 mm or smaller. Small overhead signs shall be installed in accordance with Drawings SS SP635-3.2.2 through SS SP635-3.2.6.

Sign installations shall meet the requirements of DBSS 635.32.

635.17.08 Video Detection Equipment – Video Detection equipment shall be installed in accordance with the Contract, Design and manufacturer's instructions. The cameras shall be installed on signal arms or on special combination type 2A luminaire / camera arms in accordance with Drawings SS SP635-2.3.12 and .13 The Contractor shall undertake all commissioning and set-up of the equipment under the direction of the supplier technical representative.

635.18 Service Equipment - Service equipment shall be installed in accordance with Drawings SS SP635-2.4.1 through SS SP635-2.4.16.

Service Equipment shall include electrical panels and telephone demarcation panels.

Specialized service equipment (e.g., service kiosks) shall be installed in accordance with the Design and the Contract.

Service equipment shall be securely attached to the poles.

635.19 Wiring - All wiring shall be installed in accordance with the Design or as directed by the Designer.

All conductors shall be stranded copper, RW90 XLPE insulated unless otherwise noted.

Conductor gauges (AWG) shall be as specified on the Drawings and the Design.

Shielded detector loop and telephone demarcation cables shall be 2 conductor No. 18 Ministry pre-approved product. Shielded detector loop cables shall run continuous with no splices, from the traffic controller (or other type of control equipment) to the respective detector loop conductor.

Single conductor colour coding and labelling is required and shall be as noted on Drawings SS SP635-2.5.7 and 2.5.8. Conductor colour coding shall also meet the requirements of the Canadian Electrical Code.

Where specifically noted on the Design, signal cable shall be used. Signal cable colour coding and conductor designations shall be as noted on Drawing SS SP635-2.5.9. Where signal cable is used all conductor splices shall be made in pole hand holes in accordance with Drawing SS SP635-2.5.10.

Each traffic and pedestrian signal head section and luminaire shall be wired separately to the base of the pole. A separate neutral and bond conductor shall run to each signal head and luminaire from the base of the pole. The neutral and bond conductors shall be bundled with the feeder circuits.

Luminaire conductor wiring and fusing in pole handholes shall conform to Drawing SS SP635-2.5.6.

Wiring inside junction boxes and vaults shall conform to Drawings SS SP635-2.5.1 through SS SP635-2.5.5.

Conductor splices shall be secured with solderless "Marrette" type connectors. Where the number and/or size of conductors exceed the capacity of the Marrette split bolt connectors shall be used.

All wiring shall be neatly bundled and labelled in all junction boxes, vaults, traffic controllers, handholes at pole bases, and service panels.

Sealing of connections shall be performed using one of the following methods:

a) Double dipping using 3M "Scotchcote" or approved alternate. Dipping shall be performed strictly adhering to the manufacturer's specifications.

b) Each conductor shall have a wrap of the self-holding tape (3M Linerless Rubber Splicing Tape, Steadfast 8 Ounce Splicing tape or approved alternate), then the complete splice shall be wrapped. PVC tape shall then be applied to cover the complete splice.

If conductor connections require the use of split bolts or similar style devices due to wire size, then the splice shall be completely covered with Duct Seal to form a ball over the connection. This Duct Seal shall be thick enough to prevent sharp ends of the conductors and/or points of the connector from protruding through the taped connection. Once the duct seal has been applied, the splice shall be taped with self-holding and PVC tape as described in DBSS 635.19 b).

All luminaires and signal heads shall be bonded with a No. 12 RW90 conductor. Steel junction box lids and steel vault lids shall be bonded in accordance with Drawings SS SP635-2.5.1 through SS SP635-2.5.5.

635.20 Traffic Counter Stations - Traffic counter stations shall be short-duration type or permanent type.

Short duration traffic counter stations shall be installed in accordance with Drawing SS SP635-2.6.1.

Perforated square steel tubing shall conform to DBSS 635.28.

Permanent traffic counter stations shall be installed in accordance with Drawing SS SP635-2.6.2 and 2.6.3.

The Contractor will install all pole mount cabinets. The Ministry's electrical maintenance contractor will connect all field wiring in cabinets, trouble-shoot problems in the cabinet and undertake all equipment set-up

635.21 Controllers - Type B, C, M and S traffic controller cabinets shall be installed in accordance with Drawings SS SP635-2.7.1 through SS SP635-2.7.3.

The Ministry's electrical maintenance contractor or Ministry Electrical Trades Supervisor will install all base mount traffic controller and cabinets. The Contractor will install all pole mount cabinets. The Ministry's electrical maintenance contractor or Ministry Electrical Trades Supervisor will connect all field wiring in controller cabinets, trouble-shoot problems in the cabinet, enter all signal timings, undertake modifications to existing signal phasing and timings, start-up controller and complete Ministry signal turn-on sheets.

The Contractor shall be on hand to assist with controller start-up and make any repairs to field wiring or hardware as required. The Contractor shall provide traffic control and assistance during the controller start-up.

The Contractor shall coil and label 2 m of each conductor inside the traffic controller for connections by the Ministry's electrical maintenance contractor or Ministry Electrical Trades Supervisor. The Contractor shall verify that all traffic and pedestrian signal phases are wired as shown on the Design and that all circuits are tested prior to activation. The Contractor shall verify that all signal and pedestrian phases are properly colour coded and labelled with identification tags.

635.22 Detector Loops - Detector loops shall be constructed as noted on the Design and shall be installed in accordance with Drawings SS SP635-2.8.1 through SS SP635-2.8.16. Loop check sheets shall be completed and submitted to the Ministry Representative prior to signal start-up.

635.23 Repairing Galvanized Surfaces - Any spots where the galvanized finish is damaged due to drilling, tapping, reaming or welding and any surface damage incurred during transportation and erection shall be refinished with cold galvanizing compound. Cold galvanizing compound shall be Crown No. 67007 spray type (or approved alternative). The application of cold galvanizing compound shall conform to the Manufacturer's instructions and the following:

a) The surface shall be mechanically cleaned with a wire brush or grinder and chemically cleaned to remove all welding flux, paint, grease, oil, rust, scale or other detrimental foreign matter.

b) The surface shall be absolutely dry and the ambient temperature shall be over 10°C.

c) Uniform coats shall be applied. Each coat shall be as thick as possible without causing runs on the finished surface.

635.24 Flasher Luminaires - Flasher luminaires shall be mounted on perforated square steel tubing in accordance with Drawings SS SP635-2.9.1 through SS SP635-2.9.3.

Flasher luminaires shall be mounted on poles in accordance with Drawings SS SP635-2.9.4 through SS SP635-2.9.6.

All perforated square steel tubing shall be in accordance with DBSS 635.28.

PART D - SIGNING

635.25 Overhead Signs - Overhead signs are categorized in two classes:

a) small overhead signs are 1200 mm x 900 mm or smaller;

b) large overhead sheet aluminum, plywood or extruded aluminum guide signs are larger than 1200 mm x 900 mm.

Small overhead signs are generally plywood or sheet aluminum, and are mounted on signal poles. Small overhead signs shall be installed in accordance with Drawings SS SP635-3.2.2 through SS SP635-3.2.6.

Large overhead guide signs are installed on sign poles in accordance with Drawings SS SP635-3.3.1 through SS SP635-3.3.6, unless otherwise noted. Large overhead extruded aluminum signs are installed on sign poles in accordance with Drawings SS SP635-3.3.11 through SS SP635-3.3.17, unless otherwise noted. Sign poles shall be installed in accordance with Drawings SS SP635-3.1.1 through SS SP535-3.1.19. Large overhead plywood signs are not typically used for new installations.

Sign lighting, where specified by the Ministry, shall be installed in accordance with Drawing SS SP635-3.3.18 and DBSS 635.17.03.

Overhead signs shall be securely installed on sign pole structures.

All horizontal supports and signs shall be level and vertical supports plumb.

Sign Poles shall be installed in accordance with DBSS 635.17.

Advance warning signs shall be extruded aluminum, and shall be installed in accordance with Drawings SS SP635-3.3.7 through SS SP635-3.3.10.

All signs shall be installed in accordance with DBSS 635.32.

635.26 Breakaway Sign Structures - Breakaway sign structures shall be installed in accordance with Drawings SS SP635-3.4.1 through SS SP635-3.4.12.

After concrete bases are installed, the Contractor shall survey the finished base elevations to determine the correct leg lengths. The fabricator shall cut legs to the correct lengths.

All legs and columns are to be transported in an unstressed manner.

All breakaway structures shall be installed with the legs and columns plumb, and the battens and signs level.

The Contractor shall tighten all bolts and nuts with a torque wrench to the torque specified on the Design.

Signs shall be sheet aluminum plywood or extruded aluminum, as specified.

Plywood signs shall generally be installed in accordance with Drawings SS SP635-3.4.7 and, 3.4.8 and DBSS 635.32.

Plywood signs shall be installed with wood battens on the breakaway sign columns to support the signs. Wood battens shall be 4" x 6" Douglas Fir/Larch, No. 1 Grade or pressure treated, surfaced four sides, in complete lengths without splices. Battens shall be straight and free of cracks. All wood battens shall be painted green with one coat of oil base exterior paint (General Paint No. 63-324 - Highway Green, or approved alternative) over a compatible primer (General Paint No. 32002 - Primer or approved alternative). All painting shall conform to DBSS 308 – Paint It is not required to paint pressure treated sign battens.

All pressure treated wood battens shall be pressure treated in accordance with CSA Standard 080.5 “Preservative Treatment of Posts by Pressure Processes”

Signs shall not be installed until the paint on the battens has completely dried.

Extruded aluminum signs shall be installed in accordance with Drawings SS SP635-3.4.9 through SS SP635-3.4.12 and DBSS 635.32.

Extruded aluminum signs on breakaway sign structures will not be illuminated.

All sign installations shall be in accordance with DBSS 635.32.

635.27 Wood Post Sign Structures - Wood post sign structures shall be installed in accordance with Drawings SS SP635-3.5.1 through SS SP635-3.5.9.

Excavations for wood posts shall be wide enough to allow for proper compaction of backfill around the wood posts. Wood posts shall be embedded in the ground to the depths indicated on Drawings SS SP635-3.5.1 through SS SP635-3.5.3.

Wood post excavations shall be backfilled using the excavated material except where excavated material is ruled unacceptable. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in DBSS 635.14. Unacceptable materials shall be disposed of as specified in DBSS 635.15.

Wood posts shall be Douglas Fir/Larch, No. 1 Grade or pressure treated, surfaced four sides, and shall be supplied in complete lengths without splices. Posts shall be straight and free of cracks.

All pressure treated wood posts and battens shall be pressure treated in accordance with CSA Standard 080.5 "Preservative Treatment of Posts by Pressure Processes"

All portions of wood posts that are not pressure treated installed below finished grade shall be treated with wood preservative.

Wood posts shall be installed plumb and at the proper offset and elevation.

Backfill material shall be placed in layers not exceeding 150 mm compacted thickness and shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698. Layer thickness and moisture content of the material shall be adjusted as necessary to achieve compaction.

All areas excavated and backfilled shall be restored to their original condition.

Wooden sign posts shall be completely painted, prior to installing the sign or signs, with one coat of white oil base exterior paint (General Paint No. 10010 or approved alternative) over a compatible primer (General Paint No. 32002 primer or approved alternative) for the length of the post from the ground line to the bottom of the sign. Where plywood signs are installed, the portion of the post behind the sign from the bottom of the sign to the top of the post shall be completely painted with one coat of green oil base exterior paint (General Paint No. 63-324 - Highway Green or approved alternative) over a compatible primer (General Paint No. 32002 primer or approved alternative). All painting shall meet the requirements of DBSS 308 - Paint. It is not required to paint pressure treated sign posts

Signs on single wood post structures shall be plywood or sheet aluminum. Sheet aluminum signs shall be installed in accordance with Drawing SS SP635-3.5.1 and DBSS 635.32.

Signs on multiple wood post structures shall be sheet aluminum, plywood or extruded aluminum.

Plywood signs shall be installed in accordance with Drawings SS SP635-3.5.4 and 3.5.5, and DBSS 635.32. Plywood signs shall be installed on wood battens mounted on the sign posts to support the signs. Wood battens shall be Douglas Fir/Larch, No. 1 Grade or pressure treated, surfaced four sides, and shall be supplied in complete lengths without splices. Wood battens shall be straight and free of cracks.

Wood battens shall be painted green with one coat of exterior paint (General Paint No. 63-324 - Highway Green, or approved alternative) over a compatible primer (General Paint No. 32002 primer or approved alternative). It is not required to paint pressure treated wood battens.

Extruded aluminum signs shall be installed in accordance with Drawings SS SP635-3.5.6 through SS SP635-3.5.9 and DBSS 635.32.

Signs or battens or extruded aluminum signs and angle mounting brackets shall not be installed on wood posts until the paint has completely dried.

All areas around the post shall be fully restored to their original condition.

635.28 Perforated Square Steel Sign Post Structures - Perforated Square Steel Sign Post shall be supplied in accordance with Drawings SS SP635-3.6.1 through SS SP635-6.4.

Sign posts shall be perforated square steel tubing. The perforated square steel tubing will be required in different outside dimensions as noted on the Design. The perforated square steel tubing shall be formed from 12 gauge cold rolled steel, conforming to ASTM Specification A 446, Grade A. The tubing shall be hot dipped galvanized conforming to ASTM Specification A 525, Designation G-90. Steel tubing shall have 7/16" holes on all four sides at 1" centres.

Galvanized perforated square steel tubing shall be supplied in continuous lengths, with no splices, and shall be field cut to suit the particular installation. All field cuts shall be painted with cold galvanizing compound in accordance with DBSS 635.23.

Perforated square steel tubing sign posts shall be installed on concrete bases or direct buried in accordance with Drawing SS SP635-1.1.35.

Perforated square steel tubing posts shall be installed plumb.

Signs on single perforated square steel sign posts shall be sheet aluminum or steel. Sheet aluminum and steel signs shall be installed in accordance with Drawings SS SP635-3.6.1 and 3.6.3 and DBSS 635.32.

Signs on double perforated square steel sign posts shall be plywood or sheet aluminum.

Double post plywood and sheet aluminum signs shall be installed in accordance with Drawings SS SP635-3.6.2 through SS SP635-3.6.4 and DBSS 635.32.

635.29 Round Steel Sign Post Structures - Round Steel sign posts shall be installed in accordance with Drawings SS SP635-3.8.1 through SS SP635-3.8.3.

Barrier sign posts shall be mounted on barrier stands, which shall be securely bolted to concrete roadside or concrete median barriers.

Posts and pipe sleeves shall be round Schedule 40 steel pipe conforming to ASTM Specification A 53, Grade B. The pipe shall be hot dip galvanized in accordance with CSA Specification G164 Table A. Pipe shall be field cut to lengths to suit sign mounting heights as noted on the Design or as directed by the Designer. All field cuts in galvanized steel pipes shall be coated with cold galvanizing compound in accordance with DBSS 635.23.

Signs shall be installed in accordance with DBSS 635.32.

635.30 Sign Mounting on the Side of Poles - Signs mounted on the side of steel poles shall be installed in accordance with Drawing SS SP635-3.9.1.

Signs shall be securely attached to poles.

Signs shall be installed in accordance with DBSS 635.32.

Holes drilled in galvanized steel poles shall be coated with cold galvanizing compound in accordance with DBSS 635.23.

635.31 Delineators - Delineators shall be wood, perforated steel tubing or plastic as noted on the Design.

Wood delineator posts shall be installed in accordance with Drawing SS SP635-3.10.1.

Perforated square steel tubing delineator posts shall be installed in accordance with Drawing SS SP635-3.10.2.

Where plastic delineators are specified, they shall be a Ministry preapproved type and shall be installed in accordance with the manufacturer's instructions.

Where delineators are mounted on a pole, sign post or structure, they shall be installed in accordance with Drawing SS SP635-3.10.3.

Wood delineator post excavation, backfill, type of wood, painting and installation methods shall be in accordance with DBSS 635.27.

Perforated square steel tubing shall be in accordance with DBSS 635.28.

All holes drilled in galvanized surfaces shall be coated with cold galvanizing compound in accordance with DBSS 635.23.

635.32 Signs - Small signs which are generally 1200 mm x 900 mm or smaller shall be sheet aluminum or plywood. Large signs which are generally larger than 1200 mm x 900 mm shall be plywood or extruded aluminum as specified.

All sign installations shall meet the requirements of the most current edition of the Ministry *Manual of Standard Traffic Signs and Pavement Markings* available at*;*

http://www.th.gov.bc.ca/publications/eng_publications/electrical/MoST_PM.pdf

The Contractor shall have a current copy of this document on site when installing signs.

All signs shall be completely covered with a suitable grade of polyethylene sheeting from the time they are installed until the roadway is in full operation, unless otherwise directed by the Designer. The polyethylene sheeting material shall prevent sign messages from being visible.

All signs shall be handled so as not to damage them in any way. Slip sheets between signs shall be removed carefully to avoid damage due to adhesion of the slip sheet to the sign. Signs shall be stored on end on dunnage or racks in a dry, covered area, safe from damage. Damaged signs shall be replaced by the Contractor at the Contractor's expense. Taping, screwing, nailing, gluing, bolting or stapling to sign faces or back is prohibited unless otherwise noted.

The Contractor shall confirm that all signs have the correct messaging. The Contractor shall verify that all signs are free of cracks, dents or warpage prior to installation.

Signs shall be bolted to the mounting hardware or structure as indicated on the applicable Design. Nylon washers shall be installed between the mounting bolt heads and the plywood sign face. The sign panels shall be tightened so as to eliminate sign movement but not over tightened so as to recess the bolt heads into the sign face.

Plywood signs larger than 1200 mm x 3000 mm will consist of multiple plywood panels. Each panel shall be correctly aligned so that no gaps exist between sign panels. The bottom of the sign panels shall be installed level.

Plywood exit number tabs shall be attached to plywood signs in accordance with Drawing SS SP635-3.7.1.

Extruded aluminum signs will be supplied unassembled. Aluminum sign sections shall be assembled and correctly aligned to the satisfaction of the Ministry Representative

Extruded aluminum signs are supplied in typical widths shown on Drawing SS SP635-3.3.13.

Extruded aluminum exit number tabs shall attach to extruded aluminum signs in accordance with Drawings SS SP635-3.7.2 and 3.7.3.

Any unused holes in wooden sign boards shall be sealed. Holes on the sign face shall be covered with a trimmed piece of patching material to match the colour of the sign face.

Patching material shall meet current Ministry Specifications.

All plywood, sheet aluminum and extruded aluminum signs will be identified with their sign numbers labelled on the back. Sign numbers shall be determined as directed by the Ministry Representative

The Contractor shall maintain all signs installed as part of the work within the project area for the duration of the Work. Maintenance shall be performed for the duration of the work in accordance with the following:

a) Sign maintenance shall include the straightening, replacing, repairing and cleaning of all signs installed.

b) The Contractor shall regularly inspect the signs to ensure consistent maintenance for maximum visibility.

c) Where necessary the Contractor shall clean the signs by power washing or steam cleaning using pressures not exceeding 7 MPa and temperatures not exceeding 65°C. Chemical washing of sign faces will not be accepted.

SECTION 741

FENCE CONSTRUCTION

DESCRIPTION

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

741.01 Scope - This Section covers the construction of wire fencing with wood and/or metal posts, gates, wood privacy and noise barrier fencing and, with reference to the SS SP741 drawing series, is intended to specify acceptable standards and some optional features as may be required by the Contract.

Alternative construction and methods may be called for by the Drawings, the Special Provisions or instructions of the Designer.

For SP Drawing referenced herein, such as SS SP741-01.01, see the corresponding SP Drawing in the Ministry's *Standard Specifications for Highway Construction, 2006 edition.*

MATERIALS

741.11 General - Material for fencing including wire fabric, barbed and high-tensile wire, chain link mesh and metal posts are specified in DBSS 316. Wood fence posts are specified in DBSS 909.

Chain link fabric as protection on rock slopes is covered by DBSS 207 and DBSS 316.

The types of standard wire and chain link fencing are designated in Table 741-A. Tentative requirements for High-tensile Smooth-wire Fences are included.

TABLE 741-A TYPES OF STANDARD WIRE AND CHAIN LINK FENCING

TYPE A	Special Wire Fabric Fence for use only on railway right-of-way
TYPE B	Standard Wire Fabric Fence
TYPE C	Standard Barbed Wire Fence
TYPE D	Chain Link Fence

Note: Drawing SS SP741-01.01 indicates the general requirements of Types A and B Fences, Drawing SS SP741-02.01 indicates those of Type C Fences, and Drawing SS SP741-05.01 indicates those for Type D. The requirements for Type C are acceptable to the Provincial Wildlife Branch and the B.C. Cattlemen's Association.

741.12 Standard Wire Type A, B & C Fences and Gates - Materials generally shall be in accordance with the requirements set out on Drawings SS SP741-01.01 and SS SP741-02.01, the relevant subsections of DBSS 316 and DBSS 909 and/or the Special Provisions.

Preservative treatment for protecting field cuts and notches and for making good any superficial damage to treated wood posts, braces and anchors, where permitted by the Quality Manager, shall be compatible with the original pressure treatment for application in two separate heavy coatings.

Touch-up treatment for damaged galvanizing of steel posts and braces shall be a heavy application of a zinc rich colour matched paint to CGSB Standard 1-GP-181M Specification for Ready Mixed Zinc Rich Coating.

741.12.01 Gates - Gates shall be of the prefabricated type indicated on Drawings SS SP741-04.04, SS SP741-04.05 and as specified in DBSS 316.11 and/or as specified in the Special Provisions.

741.13 High-tensile Smooth-wire Fences - Materials to be supplied in accordance with the Purchase Order will include:

- high-tensile smooth galvanized wire 2.5 mm nominal diameter
- mechanical wire splices and fasteners for high-tensile wire
- in-line high-tensile wire tensioning devices
- metal or wood droppers for high-tensile fencing.

Wood posts and braces shall be supplied by the Contractor in accordance with the requirements of DBSS 909 together with steel posts, gates and hardware, brace wire, dowels, staples and the like in accordance with the relevant subsections of DBSS 316.

741.14 Chain Link Type D Fences and Gates- Chain link mesh, steel pipe and accessory materials generally shall be in accordance with the requirements set out on Drawings SS SP741-05.01 and SS SP741-05.02, the relevant portions of DBSS 316, and/or the Special Provisions.

741.14.01 Gates - Gates shall be of the type indicated on Drawings SS SP741-05.03 and SS SP741-05.04 and/or specified by the Special Provisions, and shall be equipped with hinges (or sliding) and locking hardware.

741.15 Wood Fences - Materials used for the fabrication and installation of wood fences shall be as follows:

i) Concrete footings: Minimum 18 MPa compressive strength concrete, comprised of aggregate, sand and Portland cement (4:2:1).

ii) Lumber: Rough sawn or dressed, as specified, shall be of Western Red Cedar with:

- Wood posts and stringers of "No. 1 Structural" grade posts and framing or plank
- Boards and planks of "Quality Fencing" grade except where noise barrier fencing is specified for "Select Fencing" or plank equivalent.

Other species may be specified or approved and shall be of equivalent grades with pressure preservative treatment in accordance with CSA Standard 080 and compatible with staining requirements.

All lumber shall be in conformity with the NLGA "Standard Grading Rules for Canadian Lumber."

iii) Galvanized steel pipe: 48 mm OD for privacy fence posts shall conform to the requirements of DBSS 316.10 with weatherproof caps where open ends are not covered by wood members.

iv) Steel: Steel used as posts for noise barrier fences or as wood post supports shall be of the structural shape(s) indicated in accordance with CAS Standard G40.21M, grade 300W and where galvanized, hot dipped to the requirements of CSA G164.

v) Fastenings: Bolts generally shall conform to ASTM A 307, nuts to ASTM A 563 Grade A, plain washers to ANSI B27.2 Type A, plate washers, where required, to ASTM A 36; all galvanized according to CSA G164. Nails shall be casing headed heavy gauge of appropriate length, hot dip galvanized with deformed shank (annular, barbed or helical) for noise barrier fence plank fixing.

vi) Finish: Penetrating stain with preservative shall be of type and colour specified at least conforming to CGSB Standards 1-GP145M and 204M, to all surfaces prior to prefabrication or installation and on any cuts before final fitment.

CONSTRUCTION

741.31 Provision of Fencing - Fencing of the type(s) called for shall be carried out at the locations and as shown on the Design with the materials to the height, spacing and with accessories all in accordance with the details indicated on the Design, Standard Specifications, Special Provisions and SP Drawings.

All material shall be supplied by the Contractor.

Construction shall be carried out with all labour, tools, equipment and incidentals supplied by the Contractor, as necessary, to complete all fencing work in accordance with good work practice.

741.32 Clearing & Grading - All trees (other than any required by the Design or Contract to remain), all brush and other obstacles which interfere with the construction and maintenance of fencing and not removed by the normal clearing operations, shall be removed prior to commencing fencing work so that both sides of the fence line are free of all clearing and grubbing debris.

Minor ground undulations shall be corrected to obtain a smooth uniform grade, but appreciable grade depressions may be backfilled only with the permission of the Designer and Quality Manager.

The site shall be left in a smooth and tidy condition.

741.33 Setting Out and Connections to Existing - Fence line, as shown on the Design, generally will be along the right-of-way boundary. Where undergrowth clearing is carried out to the right-of-way boundary, the fence line will be offset 0.5 m in from the boundary.

Post installation in fill material or minimum overburden shall be carried out to the Designer's or Quality Manager's directions. Where it is not possible to drive or set wood posts to proper depth or to relocate same along the fence line, steel fence posts as specified or, where permitted, multiple wood post and brace assemblies shall be substituted.

Existing cross fences shall be connected to new with posts and braces for tensioning fencing wire in every direction in accordance with Drawings SS SP741-01.03 and SS SP741-02.02.

741.34 Standard Wire Type A, B & C Fences and Gates

741.34.01 Post Installation - Fence posts shall be driven in place with equipment acceptable to the Quality Manager, set in augered pilot holes or, where permitted by the Quality Manager, set in dug holes with necessary well tamped backfill for a firm installation and post penetration to at least the depth indicated on Drawings SS SP741-01.01 and SS SP741-02.01.

On straight alignments all posts shall be plumb. On appreciable grades, posts shall be installed perpendicular to

the slope.

On curved alignments, the posts shall be set 50 mm off plumb away from the curve centre, with a post spacing in accordance with Drawing SS SP741-04.02 and increased post lengths and bury depth for post stability where necessary.

Gate post sizes and stabilizing shall be as required by the Special Provisions and/or Drawing SS SP741-04.04.

Steel fence posts, as specified by DBSS 316.09, are required on exposed rock or rock with "minimum overburden" (as defined on Drawing SS SP741-04.01) and shall be driven and/or wedged to the full depth in a vertical drilled hole of minimum diameter or set plumb and rigid in cement and sand or fine aggregate mortar all as indicated by Drawing SS SP741-04.01.

Note: Assemblies of securely wire tied multiple wood posts/braces may be specified by the Design or permitted by the Designer especially for minor rock outcropping and unforeseen minimum overburden occurrences.

When full bury depth of wood posts is not attainable, the specified steel fence posts shall be used to the full penetration into rock, all as indicated on Drawing SS SP741-04.01.

Tops of all posts shall be set to a uniform 50 mm above the level of the top wire of fencing and, where necessary, cut to line-up in vertical uniformity Wood post tops where cut, shall be bevelled and preservative treated in two heavy applications.

Wood posts which are burred, split or otherwise damaged from the installation shall be replaced.

741.34.02 Bracing for Fence Tensioning - Braces of treated wood and twisted galvanized diagonal wire shall be installed between end posts and adjoining panel posts to stabilize the tensioning of the fencing fabric and/or barbed wire, all as indicated by Drawings SS SP741-01.03 and SS SP741-02.03. Corner and intersection assemblies shall be similarly tensioned in each fence direction.

Intermediate tensioning assemblies shall be provided within the maximum spacings indicated on Drawings SS SP741-01.02 and SS SP741-02.02 to make use of rolls of fabric etc. of standard length with a minimum of cutting and waste.

Steel corner, intersection and intermediate tensioning assemblies with steel angle posts and diagonal bracing are indicated on Drawing SS SP741-04.01.

Horizontal alignment changes over 30° with wood posts and over 15° with steel posts shall be stabilized as for corners above. Where a change is less than 30° with wood posts, a pair of line posts, set 2.4 m or 3.0 m apart, shall be stabilized with a horizontal wood brace and diagonal wire bracing both ways. For a change of less than 15° with steel posts, a steel line post at the change shall be stabilized in both directions with diagonal steel braces to adjoining line posts as indicated on Drawing SS SP741-04.01.

Similar requirements shall be maintained at each change to or within any curved fence alignment where the closer post spacing indicated by Drawing SS SP741-04.02 is not considered adequate.

Vertical alignment changes shall be stabilized, and fence requirements at ditch and gully conditions shall be as called for by Drawing SS SP741-04.03 or as the Designer may direct.

741.34.03 Fabric and Wire - Fence fabric shall normally be installed on the side of the posts away from the highway, stretched between end type assemblies and intermediate tensioning assemblies with proper equipment (tensioning directly by truck or tractor will not normally be permitted) and securely stapled in accordance with good practice to wood posts, as indicated by Drawing SS SP741-01.01, or securely wired or clipped to steel posts to permit free wire movement on line posts. Similarly, barbed wire shall be installed so as to allow it to "prestretch" before final tensioning, and shall be securely stapled to tensioning assemblies.

For fencing on a curve requiring the fabric and/or barbed wire on the highway side of the posts, all line wires shall be double stapled and the top wire shall be securely wire tied to posts in addition where the adjoining property has livestock.

Barbed wire Type C fences shall normally have wood droppers "interwoven" and securely figure-of-eight wire tied to every line wire, as indicated by Drawing SS SP741-02.01. Alternatively, proprietary galvanized sheet metal prefabricated droppers of pre-approved design for secure clip-on application may be approved.

Note: High livestock pressures may warrant Type C1 fencing and nursery livestock enclosures Type C1 or B fencing. Where such fences, 1200 mm and higher, are in wildlife sensitive areas, the top strand may be specified smooth (barbless) as Type C2 or B1 fences.

741.34.04 Gates - Hardware, of design to permit the gate to operate correctly, shall be securely attached to prevent the easy removal of the gate and hardware. Hinges shall be installed to permit the gate to swing back one-way against

the fence. Locking hardware shall be of the type specified.

741.35 High-tensile Smooth-wire Fences - For rangeland and other locations with favourable terrain, soil conditions and fence alignment at the locations indicated on Design, construct high-tensile smooth-wire fencing between rigid end, gates, corner and any necessary intermediate tensioning assemblies in general accordance with Drawings SS SP741-03.01 through SS SP741-03.03.

741.35.01 Fence Style - Fence style for wildlife crossing areas shall comprise 5 wires, the bottom wire for installation at 250 mm ± 50 mm above grade at and between posts, the next two wires at 200 mm spacing and the top two 225 mm for a total height of 1100 mm, as indicated by Drawing SS SP741-03.01 for HT Fences. Elsewhere, and for nursery livestock enclosures and other areas of high livestock pressure, a 6-wire fence may be specified, having the bottom wire 200 mm ± 50 mm above grade at and between posts with the remaining wires at 200 mm spacing for a total height of 1200 mm indicated as HT-1 Fences.

741.35.02 Post and Brace Installation - Fence posts shall be driven in place with equipment acceptable to the Quality Manager and set out in general accordance with the recommendations of the B.C. Ministry of Agriculture and Food (BCMAF) Publication #ISBN 0-7719-9824-4, good local trade practice and to the Quality Manager's direction.

Terminal (end and gate) tensioning assemblies and any intermediate assemblies shall be "single" assemblies and at the spacings and post bury depths indicated by Drawing SS SP741-03.02 (or such greater depths and/or decreased spacings as may be directed by Designer) with "double" assemblies used with scant bury depths all to form rigid, stable and accurately aligned assemblies for tensioning line wires.

Line posts 100 mm minimum diameter driven to a depth of not less than 650 mm in firm soil, or as otherwise required by Drawing SS SP741-03.01, or directed, shall be spaced up to a maximum of 15 m where permitted, but normally at 10 m spacing and such closer spacing at changes in horizontal and vertical alignments to keep the bottom wire at the regular specified ground clearance.

741.35.03 Wire and Dropper Installation - Wires shall be installed on the side of the fence posts away from the highway except on such curves requiring the wires on the highway side so as to bear on the posts.

Stapling doubled at curves, etc. and mechanical fasteners at tied-off ends and splices (where permitted) shall be to BCMAF recommended practice.

With supplied tensioning devices installed in each line, all wires shall be individually tensioned in stages and after "wire-set" to 1.33 kN (300#) at completion.

Droppers, as supplied, shall be installed and securely clipped to all line wires at 3 to 4 m spacing.

741.36 Chain Link Type D Fences and Gates

741.36.01 Post and Rail Installation

i) All terminal posts (posts at ends, gates, corners and intersections), all line posts and any intermediate tensioning posts shall be set plumb into concrete footings in augered or dug holes to the depths and regular spacing all as shown on Drawing SS SP741-05.01, or as otherwise specified.

ii) Gate post sizes and stabilizing shall be as required by the Special Provisions and/or Drawing SS SP741-05.03.

iii) On exposed rock, posts shall be set without concrete footings to full depth and fully grouted in holes, 25 mm greater than the pipe diameter, drilled to a depth into solid rock of half the depth of pipe bury to that indicated for concrete footings on Drawing SS SP741-05.01 (i.e. c/2 and f/2).

Grout shall be non-shrink cement and sand mortar. Standard length posts may be used where the overburden depth is such that the depth of bury into solid rock is less than that specified above, provided normal diameter concrete footings of the diameter indicated on Drawing SS SP741-05.01 are formed from solid rock to grade level. Similarly, form concrete footings for posts set in loose or friable rock. Sleeves shall be provided to form holes similar in diameter and depth to above for casting into concrete where shown or specified.

iv) Tops of all posts shall be set or cut for an even height of top rail, which shall form a continuous brace and mesh support between terminal posts and any intermediate tensioning posts. Top rails shall pass through line post caps and be joined in the length with internal sleeves to allow expansion and contraction. All posts shall be fitted with appropriate weathertight caps securely fixed.

741.36.02 Bracing for Fence Tensioning - Bottom tension wire shall be securely fixed taut and sag free to terminal posts and any intermediate tensioning posts. Similarly, provide top tension wire, when specified, in place of top rail to pass through line post tops except on uneven terrain where the height of the top tension wire shall be such that it is secured to all line posts and the mesh within the top

300 mm of the mesh.

Terminal posts, where more than 10 m apart in any fence run, shall have horizontal pipe braces to adjoining line posts. Diagonal pipe braces may be called for where soil conditions warrant, especially where fencing without top rails is specified, but no pipe bracing is normally required for residential height fences.

Intermediate tensioning assemblies shall be provided where terminal posts are more than 150 m apart, and at any subsequent 150 m maximum spacing, to consist of a straining post (to full height of fence where barbed wire on extension arms is specified) with horizontal pipe braces at the mid to two-thirds height above grade to adjoining line posts each way for the discontinuity of top rail, tension and/or barbed wire and mesh; provide similar tensioning assembly at abrupt vertical alignment changes.

Horizontal alignment changes where abrupt shall be considered as corners.

At changes in horizontal alignment and to curved fence alignments, where the top rail can be continuous by accurately bending to proper curvature without damage to the galvanized coating, a pair of line posts shall be stabilized and tensioned as follows:

- A horizontal pipe brace shall be securely fixed to adjoining line posts at the two-thirds height above grade.
- Crossed diagonal wire braces shall each be two strands of 3.5 mm nominal diameter galvanized wire securely fixed to the brace band fixing of the horizontal brace and to a brace band 100 mm above grade.
- Each pair of wires shall be twisted taut to mutually stabilize the assembly with the tensioning battens left in place.

Similar requirements shall be maintained within any curved fence alignment over 30 m in length.

Fencing where continuous over a creek or ditch shall be stabilized with a horizontal pipe brace and diagonal brace wires as specified above, and the bottom tension wire shall be wire tied to a similar pipe brace. In addition, in-fill under where specified or required, shall be a hanging or rigidly fixed bent pipe of brace diameter hung or in-filled with closely spaced barbed wire or suitable fence fabric (similar to that indicated on Drawing SP741-04.03), all to the Designer's direction.

741.36.03 Fabric and Wire - Chain link fencing mesh shall be stretched between terminal posts and any intermediate tensioning posts with proper equipment (tensioning directly by truck or tractor will not normally be permitted), and secured with tension bars and bands, tie wire and clips all in accordance with the requirements shown on Drawing SS SP741.05.02. The mesh shall be installed on the highway side of the fence posts, or as otherwise specified or directed, and joints in the length shall be made by weaving the meshes together with a single wire picket to form a neat continuous fabric mesh.

Security barbed wire, where specified, shall be installed in the slots of all extension arms and secured to extended height terminal and any intermediate tensioning posts taut and free of sag.

741.36.04 Gates - Hardware, of design to permit the gate to operate correctly, shall be securely attached to prevent the easy removal of the gate and hardware. Hinged gates shall be installed to permit the gate to swing back one-way against the fence.

741.37 Wood Fences

741.37.01 Privacy Fences - Galvanized pipe posts shall be set plumb in concrete footings in augered or dug holes to the depth and regular spacing all as shown on Drawing SS SP741-06.02.

Wood fixing members for fence panels shall be securely nailed, "II-shaped" frames fixed and supported with 9.5 mm (3/8") galvanized bolts and nuts to pipe posts all as indicated on Drawing SS SP741-06.01.

Metal fixing members for fence panels, when specified, shall be of named or otherwise pre-approved proprietary brand or detailed formed metal channel and strap, bolted on around pipe posts and securely machine screwed and tapped at the correct levels all as indicated on Drawing SS SP741-06.03.

Fence panels of boards and stringers prestained, as specified, shall be prefabricated to the required design, as indicated by the Design, Special Provisions and/or Drawings SS SP741-06.01 through SS SP741-06.03, and shall be accurately fitted and securely nailed to fixing members on steel posts to form uniformly level and/or stepped fencing.

741.37.02 Noise Barrier Fences - Posts of indicated type, size and length shall be set plumb in concrete footings in augered or dug holes to the depth and regular spacing all as shown on the Design.

Stringers, to bear full width of post at each end of "alternating" or "zig zag" (on plan) panels, shall be securely bolted with large washers behind head and nut.

Planking shall be firmly secured to stringers with deformed shank nails to form close butted and battened or tightly overlapped vertical planked panels. All gaps, especially at bottom, shall not exceed 1% of the total area of the fencing. Ends of panels shall be closed over the gap between stringer and post with return plank material.

SECTION 751

TOPSOIL AND LANDSCAPE GRADING

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

751.01 Scope - This Section refers to those portions of work that are unique to the supply and placement of topsoil and subsequent finish grading. In this Section, the term "topsoil" is used to identify either:

a) On-Site topsoil: material stockpiled for use, or

b) Imported topsoil, or

c) Manufactured topsoil (Growing Medium).

751.02 Related Work - DBSS 754, Planting of Trees, Shrubs and Ground Covers; DBSS 757, Revegetation Seeding.

751.03 References - Canadian System of Soil Classification, Canada Fertilizer Act, Canadian National Master Specification.

751.04 Topsoil Supplied by the Contractor - The Contractor shall advise the Ministry Representative of the sources of topsoil not less than seven days before any is used in the work.

The Contractor shall acquire a soil analysis from an accredited soil testing laboratory, to verify that supplied material is within the requirements indicated.

MATERIALS

751.11 General - In this Section, a range of measurable physical and chemical properties are set out as being acceptable in a topsoil. Compliance with this Section shall be determined by testing for those properties. When imported or on-Site topsoil is used, it shall be tested and modified as necessary by a mixture of other components to bring its properties to within the range set in DBSS 751.16, or as stated in the Contract. Topsoil shall not be prepared or handled in an excessively wet or frozen condition, or in any manner in which structure is adversely affected.

751.12 Topsoil Types - Three topsoil types are described in DBSS 751.13 through DBSS 751.15. Regardless of origin, all types shall conform to DBSS 751.16.

751.13 On-Site Topsoil - On-Site topsoil may be used provided that it meets the standard set for imported topsoil and can be modified to meet the requirements set out for the specified topsoil. On-Site topsoil shall be defined as the existing "A" horizon containing accumulated organic matter. On-Site topsoil shall be tested prior to stockpiling.

751.14 Imported Topsoil - Imported topsoil shall be of a sandy loam or loamy sand texture (no less than 50% sand by weight) containing between 4% and 15% organic matter (dry weight basis).

Imported topsoil shall be free of propagules of plant species designated as noxious weeds under the BC Weed Control Act and Regulation and other perennial, weedy plants such as couch-grass, horsetail, broom, Japanese Knotweed and blackberry species.

At least 80% of imported topsoil shall pass a Tyler #10 sieve after appropriate crushing of structural units using accepted laboratory test methods.

751.15 Manufactured Topsoil - Manufactured topsoil is any soil or growing medium mixture whose chemical and physical properties fall within ranges required by this Section for a particular application.

Manufactured topsoil shall conform to DBSS 751.16 or as specified in the Contract.

751.16 Requirements For Topsoil - Commercial processing and mixing of topsoil components shall be done thoroughly by a mechanized screening process. No hand mixing shall occur. The resulting product shall be a homogenous mixture having the required properties throughout.

The general amendment of both natural topsoils and manufactured topsoils by mixing in situ with rototill cultivation equipment after placement is an acceptable method. Fertilizers typically used at time of seeding or planting will be surface applied and incorporated as described in DBSS 751.33, Applying Fertilizers.

The Contractor shall also require the laboratory to include recommendations for incorporating fertilizers and other amendments into the topsoil as needed for plant establishment and maintenance, and as they specifically relate to:

- grassed areas
- ground covers, shrubs and trees
- container or planter box installations, and to site conditions and season of planting.

Refer to Table 751-A for required properties of growing medium for different applications.

Other specific requirements are as follows:

Fertility :

- Nitrogen - total nitrogen shall be 0.2% to 0.6% by weight.
- Phosphorus - available phosphorus shall be 20 to 100 ppm.
- Potassium - available potassium shall be 50 to 250 ppm.
- Boron - concentration in saturation extract shall not exceed 1 ppm

General:

- Acidity - in accordance with Table 751-A. Maximum of 0.5 kg/m2 of dolomite lime to achieve the required pH level.
- Salinity - saturation extract conductivity shall not exceed 3.0 mmho/cm at 25 degrees C. Sodium absorption ration (SAR) as calculated from analysis of saturated extract shall not exceed 8.0.
- C/N ratio - carbon to nitrogen ration shall not exceed 40:1.
- Texture - in accordance with Table 751-A.
- Organic content - in accordance with Table 751-A
- Cedar or redwood sawdust shall not be present in the topsoil.
- Soil shall be virtually free from subsoil, wood including woody plant parts, weeds, toxic materials, stones over 30 mm, and foreign objects.
- Drainage - in accordance with Table 751-A. Drainage of growing medium can be measured only after growing medium is in place. Mixing and handling of growing medium shall be done in such a manner that the minimum saturated hydraulic conductivity indicated is achieved.

751.17 Soil Amendments - The following amendments shall be added to the topsoil as required.

751.17.01 Peat Moss - Peat moss shall be Horticultural grade, partially decomposed fibrous or cellular stems and leaves of sphagnum mosses with a texture varying from porous to spongy fibrous, fairly elastic and substantially homogeneous with pH value not less than 3.5 and not greater than 6.5, medium to coarse shredded, suitable for horticultural purposes.

751.17.02 Sand - Sand shall be clean river pump sand or alternative suitable source, free of impurities, chemical or organic matter.

Particle size in sand shall be as follows:

- 95 - 100% passing a 4.75 mm sieve;
- 0 - 40% passing a 0.600 mm sieve;
- 0 - 5% passing a 0.075 mm sieve.

751.17.03 Manure and Compost - Manure shall be well-rotted farm animal manure or mushroom manure, rotted to the extent that the material is crumbly. Manure shall be free from weed seeds, rocks, sticks, rubble and shall contain not more than 40% composted sawdust, straw or shavings. Manure shall be free of propagules of plant species designated as noxious weeds under the BC Weed Control Act & Regulation and other perennial, weedy plants such as couch-grass and horsetail.

Commercial compost shall be free from all weed seeds, coliform, pathogens and chemical or toxic contaminants. Physical contaminants such as rocks, plastic, metal or glass shall be less than 0.5%. Compost shall not be derived from or contain processed municipal sewage sludge, unless such product is authorized for use by the Ministry of Environment, and meets all local regulations and approvals.

751.17.04 Wood Residuals - Raw sawdust and woodwaste are not acceptable components of topsoil. Wood residuals used as a component of topsoil, compost, farm animal manure or mushroom manure are acceptable provided they are rotted and the total Carbon to total Nitrogen ratio for the topsoil is a maximum of 40:1.

751.18 Fertilizers - Fertilizers shall be standard commercial brands, meeting the requirements of the Canada Fertilizer Act.

All fertilizers shall be in granular, pelleted or prill form, and shall be dry, free-flowing and free from lumps.

Fertilizers shall be packed in standard waterproof containers, clearly marked with the name of the manufacturer, weight and guaranteed analysis.

All fertilizer shall be stored in a weatherproof storage place and in such a manner that it will stay dry and its effectiveness will not be impaired.

The types, formulations, and rates of application for fertilizers and liming agents to topsoil supplied by the Contractor shall be as recommended by a laboratory soil specialist on the basis of tests of the topsoil.

751.19 Fill Material - Fill Material shall not be toxic to plant and animal life in part or in concentration.

TABLE 751-A PROPERTIES OF TOPSOIL FOR DIFFERENT APPLICATIONS

PROPERTIES	Low Traffic Lawn Areas Trees and Large Shrubs	Planting Areas, Planters, Shrub and Groundcover Areas
TEXTURE: Particle size classes by the Canadian system of soil Classification	Percent of Dry Weight Mineral Fraction (%)	
Gravel greater than 2 mm, less than 30 mm	0 - 10	0
Sand greater than 0.05 mm, less than 2 mm	50 - 70	50 – 70
Silt & Clay combined	Maximum 25%	Maximum 25%
ACIDITY (pH)	6.0 - 7.0	4.5 - 6.5
DRAINAGE: Minimum saturated hydraulic conductivity (cm/hr) in place	2.0	2.0
ORGANIC CONTENT: Percent of Dry Weight (%)	5 - 10	14 – 20

CONSTRUCTION

751.31 Area Preparation

751.31.01 Stripping of Topsoil - Existing top soil material shall be stripped and removed to stockpile(s) within the project area, kept properly drained, and maintained in a neat and presentable condition free of spoil and subsoil material for subsequent spreading on prepared rough graded areas.

The storage of topsoil shall not interfere with the effective utilization of a granular source or borrow pits.

751.31.02 Preparation of Landscape Area Subgrade - This Section applies only to grading of landscaped areas outside the roadway prism.

Rough grading shall be carried out by necessary cutting and filling work to produce the lines and grades shown on the Design and allowing for the stipulated new topsoil thickness.

Surplus excavated material shall utilized on-Site in accordance with the Design or shall be removed from the Site and disposed of at the Contractor's expense

The landscape area subgrade shall be prepared to a consistent 80 - 85% Proctor density.

Soft and unstable areas below the landscape area subgrade that cannot be compacted to this standard shall be excavated and filled with suitable fill material, except in locations where special environmental conditions have been identified. In such cases, appropriate alternative solutions shall be approved by the Ministry Representative and environmental agencies as required and carried out.

Debris, roots, branches, stones, building material, contaminated subsoil, visible weeds and anything else that may interfere with the proper growth and development of the planned finished landscaping shall be removed.

Fill materials shall be placed so as to achieve stability. This may necessitate placing in lifts of 225 mm and compacting each layer to 80 - 85% standard proctor density.

Grade transitions of landscape area subgrade should be smooth and even and shall be such that ponding cannot occur on the landscape area subgrade surface.

Existing land forms shall be warped and blended into the landscaped areas with a minimum of visual disharmony.

751.31.03 Fine Grading - Areas requiring topsoil shall be fine graded by raking out spoil material and debris such as rocks, asphalt and concrete over 50 mm in diameter.

Naturalized areas not requiring topsoil shall be similarly cleaned, raked and manicured.

751.31.04 Scarifying - All landscape area subgrade shall be scarified to a minimum depth of 150 mm immediately before placing topsoil.

751.31.05 Cleanup - All unsuitable material and inorganic debris shall be removed from the project area by the Contractor.

All surplus or unsuitable organic waste and debris shall be removed from the Site unless its complete burning is approved by the Ministry Representative in compliance with the B.C. open burning regulation.

751.32 Placing Topsoil - The landscape area subgrade shall be appropriately prepared prior to placing of topsoil

The topsoil shall be:

- placed over the prepared landscape area subgrade and shall be allowed to settle or be compacted by light rolling such that it is firm against deep footprints and shall not be compacted more than necessary to meet this requirement;
- moist (25% to 75% of field capacity) but not wet when placed, and shall not be handled if frozen or wet such that its structure will be altered;
- manually spread around trees, shrubs and obstacles;
- evenly spread to a depth which after settlement or light compaction will be that shown in the Design .
- During hauling and spreading, the paved roadway and other finished surfaces including subgrade under future base courses shall be kept clean and free of all topsoil.

751.33 Applying Fertilizers - Fertilizers shall be added to bring topsoil fertility within the ranges set out in this Section or as specified in the Design.

Manufactured topsoils and processed imported topsoils will typically have fertilizers and amendments incorporated at the time of mixing and screening, while other topsoils will receive in situ amendment.

Fertilizers normally applied at the time of seeding and planting are specified under the appropriate sections. These fertilizers are generally supplemental to the base fertility requirements outlined for topsoil and are applied after topsoil is in place.

Surface applied fertilizers shall be evenly spread over the topsoil with a suitable mechanical spreader and fully incorporated to a minimum depth of 50 mm.

Lime used for top dressing shall be thoroughly cultivated into the top 100 mm of topsoil.

751.34 Finish Grading - After placing the topsoil to the finish elevations and contours required, the grade shall be finished to a high standard, to the grades shown on the Design with a smooth and even surface. Rough spots and low areas shall be eliminated to ensure positive surface drainage, and the surface shall be left smooth, uniform, free of debris and firm enough to resist deep footprinting.

Topsoil placed in traffic islands and medians shall be crowned for drainage, as shown on SS Drawing SP751-01.

SECTION 754

PLANTING OF TREES, SHRUBS, AND GROUND COVERS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

754.01 Scope - This Section refers to those portions of the work that are unique to the supply and planting of trees, shrubs and ground covers, including seeded and sodded areas that are not designated for treatment under DBSS 757 "Revegetation Seeding". This Section must be referenced and interpreted simultaneously with all other Sections pertinent to the works described herein.

754.02 Related Work – DBSS 751, Topsoil and Landscape Grading; DBSS 757, Revegetation Seeding; .

754.03 References - Canadian Standards For Nursery Stock (Canadian Nursery Trades Association), BC Weed Control Act & Regulations, Canada Seed Act, British Columbia Standard for Turfgrass Sod.

754.04 Warranty

754.04.01 The Contractor shall warranty all materials and quality of work for a period of one full year from the Actual Completion Date.

754.04.02 The warranty includes replacing all plants as determined by the Ministry Representative which are found dead or failing during the entire warranty period. Replacements shall be made immediately unless otherwise directed by the Ministry Representative, and conditions of the warranty shall apply to all replacement seeding for one full growing season.

Approval of plant material at the source does not preclude rejection of non-conforming stock on the Site prior to, or after planting.

The warranty shall not apply to plants and planted areas damaged after Acceptance by causes beyond the Contractor's control, such as vandalism, "acts of God", "excessive wear and tear", or abuse.

MATERIALS

754.11 Source Quality Control - The Contractor shall notify the Ministry Representative of the sources of plant materials to be supplied at least two weeks prior to commencing the landscape work. All non-seed plant materials shall be made available for inspection at the nursery by the Quality Manager or Ministry Representative upon at least three days notice. Field grown material is not to be dug prior to inspection.

Approval of plant material at the source does not preclude rejection of non-conforming stock on the Site prior to, or after planting.

Imported plant material shall be accompanied by all necessary permits and import licences, and shall conform to federal and provincial regulations.

It shall be the Contractor's responsibility to ensure that all regulations pertaining to the import of plant materials or their movement to or from a particular region of the province are adhered to, and all inspection certificates required by the Ministry of Agriculture and Lands are completed to the satisfaction of that Ministry.

754.12 Transporting Plants to Site

754.12.01 Dormant Period

Deciduous: Bare Root Stock (only in dormant period): Adequate protection shall be given in order to preserve moisture around the root system. For short transit period, four hours or less, maximum temperature in the truck shall not be above 20°C. In all cases, at all times, roots should be protected from frost, wind and sun (e.g., a closed van with wet straw or other suitable packing material protecting the roots). The temperature shall be maintained as uniformly as possible by mechanical means, or in any event to prevent frost damage to roots. The appropriate temperature range shall be between 1°C and 10°C.

Evergreens: It is recommended that root balls not be subjected to freezing temperatures below -5°C for a period longer than four hours. Plants shall be protected to prevent desiccation by wind and sun.

754.12.02 Non-Dormant period - Deciduous and Evergreen plant material shall be transported in a closed van or well-covered truck with a tarp or similar material in order to protect the leaves or needles from windburn. When in transit, with protection of a tarp cover only, it is recommended that foliage be sprayed with an antidesiccant. For the above material in transit for more than three days, it shall be unloaded and then stored away from direct sun for 24 hours to avoid leaf burning.

Sod shall be protected during transportation to prevent drying out and shall arrive at the Site in a fresh and healthy condition.

754.13 Unloading and Handling Procedures - Plants shall be carefully handled to minimize disturbance to root systems

and damage to stems and branches. Plants shall not be dropped to the ground when unloading. Plant material that is mishandled and showing evidence of damage to root balls, or undue damage and breakage to top growth will be rejected.

Plants are to be kept in a moist condition at all times. All plants shall be well protected against physical damage and desiccation until they are planted on the Site.

B.R. (Bare Roots) Stock: Roots shall be covered and protected immediately from frost, sun and wind.

Stock in Pots/Containers: Shall be handled as much as possible by pots only in order to reduce breakage of branches/leaves.

Balled & Burlapped (B & B) Material: Plants shall be handled by holding the root ball and supporting the stem to minimize disturbance to the root ball and damage to stems and branches.

Material in Wire Basket: Specimen trees shall be lifted and supported by the wire basket and not by the trunk.

All plants shall be promptly unloaded and their condition checked immediately upon arrival. Watering shall be provided as required and necessary pruning of minor breakage on branches performed.

Protection Against Stem and Branch Damage - During loading, transportation, off-loading and planting, all trees shall be protected against damage to stems and branches. This applies particularly to larger wire-basketed trees.

Bark shall be protected against chafing and cuts by providing a wrapping of cardboard, sackcloth or other material as appropriate and when required.

754.14 Storage

754.14.01 Storage During Growing Season - All plants in containers, balled and burlapped, or in wire basket, if not planted within three days, shall be stored in an upright position in an area providing even light and offering protection from wind and sun scald. Enough space shall be provided between plants so that light reaches all around to the bottom of the plant in order to avoid leaf burning when planted out.

Sod: Sod shall be installed as soon as possible after delivery. If there is a delay of more than 24 hours, the sod shall be properly stored and kept moist and cool until it is placed. Sod shall not be stacked more than three levels high while being stored.

Balled and Burlapped Material: Special attention shall be given to the root ball, and unless weather is rainy or cool, root balls shall be protected by covering with material suitable to protect them from drying out (e.g. sawdust, peat moss, topsoil). Plants intended to be planted in the open shall not be kept stored in a building or any area of low light intensity for a period exceeding seven days. All plants shall be kept well watered and protected from heat and frost.

Containerized Plants: In extreme weather, freezing or high dry heat, the containers shall be buried in a protective medium. Plants intended to be planted in the open shall not be kept stored in a building, truck or any area of low light intensity for a period exceeding seven days during the growing season.

754.14.02 Storage During Dormant Period - Plants shall be cared for according to each plant's requirement for winter protection, and according to geographical location.

754.15 Plant Material

754.15.01 Species - Selection of species shall be as specified Contract and the Design. Every effort shall be made by the Contractor to obtain the plant material specified. Where evidence is submitted that a Contract-specified plant cannot be obtained, substitutions in kind, size and grade shall be made upon written approval by the Ministry Manager.

754.15.02 Origin and Requirements - All plant material shall be nursery grown stock or approved collected native plants unless specified otherwise. All nursery grown plants shall, as a minimum, comply with the Canadian Nursery Trades Association Specification “Canadian Standards for Nursery Stock” with respect to sizing, grading and quality.

Plants shall be true to name, type and form and shall be representative of their species and variety.

All plants shall be sturdy stock, with tree and shrub heights proportional to trunk caliper, overall plant width and size of root ball,. Plants that are weak and thin, and those showing effects of being grown too closely together or poorly maintained, will not be accepted.

Plants shall be vigorous and healthy with normal, well developed branches and good fibrous root systems and be free from decay, physical injury, disease and insect damage and infestation.

In particular, conifers shall have a healthy, single leader with well shaped whorls of vigorous, newly growing branches and shall exhibit natural growth habit characteristic of the species and variety. Trees sheared as Christmas trees are not acceptable.

The root balls of dug material and the soil of containerized plants shall be free from pernicious perennial weeds. All balled and burlapped plant stock shall be supplied in biodegradable root ball sacking.

The search area for plants shall include, but not necessarily be limited to, the provinces of British Columbia, Alberta and the states of Washington, Oregon, California (northern portion), Idaho and Montana. All plant material being sourced from milder areas shall be properly "hardened off" prior to shipping and planting.

754.15.03 Nursery Grown Stock - All plants specified "Container" shall be grown for the length of time necessary to permit the roots to fill and hold the soil within the container, as required by the Canadian Standards for Nursery Stock.

Similarly, all field dug material will show evidence of having been root pruned to encourage fibrous root system development and resulting in root balls that retain their integrity during handling.

Forestry seedlings shall be supplied and handled in accordance with the requirements of the Silvicultural Manual, Ministry of Forests, Province of B.C., 1993. This manual can be found on the Internet at: http://www.for.gov.bc.ca/hfp/pubs/silvman/index.htm

All plant material shall be appropriately identified and individually labelled with weatherproof tags. In the case of small containerized plants such as ground covers and bundled bare root seedlings, which are supplied in large numbers, labelling shall be limited to identification of group lots.

754.15.04 Collected Plant Stock - The Contractor shall provide either permits or verification that permission was obtained for collecting native and/or introduced plant materials. Information shall be supplied on where, when and how collection was made.

All collected plant material shall have been grown and maintained in a nursery environment for a minimum of one growing season, unless, at the sole discretion of the Ministry Representative, certain species are approved for a lesser period of time. As for nursery grown stock, collected plants shall be held long enough prior to planting to allow roots to fill the container or the dug root ball and to retain the soil within.

Collected plant stock shall be appropriately labeled with weather proof tags for easy identification at the job site.

754.15.05 Seed - Seed quality and type shall conform to that specified in the Contract and Design

All other requirements are referenced in DBSS 757 Revegetation Seeding.

754.15.06 Sod - Sod shall be nursery grown, true to type and conform to the British Columbia Standard for Turfgrass Sod, and the general requirements of the Canadian Standards For Nursery Stock.

Sod grade shall be as specified in the Contract and Design.

754.16 Water - Water shall be clean and potable and shall be supplied by the Contractor.

754.17 Fertilizer - Fertilizer shall comply with the provisions of the Canada Fertilizers Act and Fertilizer Regulations. Fertilizer shall be supplied to the specifications in the Special Provisions.

754.18 Bark Mulch - Bark mulch shall be sized 25 mm and minus, Douglas Fir or Hemlock bark chips and fines, or a combination of both types and of the quality used for decorative landscape mulching purposes. It should be free of chunks and sticks, dark brown in colour and free of all soil, stones, roots or other extraneous matter.

754.19 Backfill Topsoil - Backfill topsoil for planting operations shall conform to the requirements of DBSS 751 - Topsoil and Landscape Grading.

754.20 Other Materials - When required, various other materials such as soil amendments, erosion control products, hydraulic mulches, etc. shall be supplied to the specifications in the Contract and Design.

CONSTRUCTION

754.31 Scheduling - Work shall be scheduled to meet the milestone dates provided in the Contract, and to ensure its execution meets the requirements of living plant material.

The work shall be co-ordinated with the schedule of other trades, and be well integrated with other specific requirements such as Sediment and Drainage Management Drawings, which may be provided for any given project.

754.32 Preplanting Operations - The Contractor shall ensure that all requirements of DBSS 754.11 through DBSS 754.15 have been met and that any minor damage to plant stock is taken care of through appropriate pruning or other measure. When conditions require it, the Contractor shall apply anti-desiccant to large conifers and deciduous trees that are in leaf. Application will be in accordance with the manufacturer's instructions for the particular product.

754.33 Location of Planting - Locations, quantities and spacing of trees, shrubs, vines and groundcovers as shown on the Design shall be considered approximate and may

require adjustment by the Contractor to meet field conditions. If underground obstructions are uncovered they shall be reported to Designer for resolution.

754.34 Area Preparation

754.34.01 Finish Grade Preparation - The Contractor shall verify that grades are correct.

754.34.02 Planting Beds and Grass Areas - Prepare planting beds and grass areas in accordance with DBSS 751 - Topsoil and Landscape Grading.

754.34.03 Planting Holes - Planting holes shall be dug in accordance with the specific requirements described below. The bottom of planting holes shall be scarified and loosened to a depth of 100 mm prior to placement of plants and backfill soil.

Subsoil, rocks, roots and extraneous material shall be removed from excavated material that will be used as planting backfill soil. Unsuitable or excess material shall be disposed of.

Holes dug by a mechanical tree spade shall have their sides scarified to loosen any compaction glazing caused by the blades. Planting holes shall be tested by filling with water. Inadequate drainage conditions permitting the retention of water in planting pits for more than 12 hours shall be rectified by the Contractor before proceeding with the work.

i) Free Draining Sub-Grade: Where the subgrade and existing native surface soils are of good drainage and of a non-compacted nature, planting holes shall be excavated and prepared to allow the following depth of topsoil backfill underneath and around the root ball:

- For plants up to and including 27 cm (#5) pot size - not less than 150 mm .
- For plants larger than 27 cm (#5) pot- size not less than 300 mm

ii) Poor Draining Sub Grade: Where the subgrade and existing native surface soils are of poor texture and conditions are generally compacted, planting holes shall be excavated and prepared to allow the following depth of topsoil backfill underneath and around the root ball:

- For plants up to and including 27 cm (#5) pot size not less than 300 mm
- For plants up to 45 cm pot size - not less than 450 mm
- For tree root balls larger than 45 cm - not less than 600mm

If severely compacted conditions are encountered, and surface or ground water entering the excavations does not drain, the Contractor shall correct the problem by;

- providing a means of sub-surface drainage
- utilizing elevated planting techniques where some of the planting soil will be placed into a partial excavation and the remainder on the surface to meet the depth requirement for growing medium, or
- considering alternate planting sites.

754.35 Time of Planting - All planting operations shall be performed during the normal planting season for each type of material, and within the milestone dates provided in the Contract, unless otherwise authorized in writing. During the specified timeframe, plant operations shall, as far as practicable, take advantage of soil and weather conditions favourable to the work.

Planting into frozen ground is not acceptable.

754.36 Planting Procedures - Trees and Shrubs

General Procedure - Plants shall be installed so that after settlement they will be at the same planting depth they were at in the field or in containers. The soil mark on the stem is an indication of this, and it shall be flush with the finished level allowing for settling of the topsoil after planting and settlement. The entire root ball shall be covered with growing medium.

Once the bottom of the planting hole is scarified and the initial lift of backfill topsoil is placed, the holes shall be pre-watered and allowed to drain prior to installation of plant material.

Plants shall be set plumb in the planting beds or in the centre of the pits except where the plant's character requires variation from this.

The growing medium shall be placed in layers around the roots or ball, preferably by hand. Each layer shall be firmed to eliminate air void and ensure good soil contact with the roots. The process shall be carried out carefully to avoid injuring the roots or ball, or disturbing the position of the plant.

Trees requiring staking shall have support stakes placed carefully between the roots before backfilling. Specifications for tree supports are described in DBSS 754.

After the planting hole is filled with soil to ground level, the plant shall receive a thorough watering. A final backfill layer shall be applied to form a saucer-like berm around the circumference of the planting hole in order to catch and hold rainwater. This rain basin shall be maintained until final acceptance of the work.

Once planting and mulching is complete, the Site shall be cleaned of all excess soil, rock and debris.

Specific Planting Requirements:

Bare Root Stock - The roots of bare root plant material shall be soaked in water prior to planting. During installation, the plant roots shall be evenly spread out over a cone of soil in the bottom of the hole, and the plant supported to the correct depth as backfilling takes place. The plant shall be gently shaken in a vertical motion to ensure that soil particles sift into the root system and establish close contact with the roots

Container Stock - Non-perishable, impervious containers such as plastic pots and tubs shall be removed from plants before planting. Once plants are removed from these containers, root systems shall not be disturbed with the exception of unraveling any roots starting to spiral around the root ball.

Bio-degradable containers such as peat or paper fibre pots shall not be removed before planting, but shall be thoroughly soaked with water prior to placement in the holes. This will ensure that containers absorb subsequent watering and not repel it due to the presence of a dry barrier. The rim of such containers shall be not be exposed to the air, and when necessary, shall be removed after planting.

Balled and Burlapped Stock - When backfill soil is placed to approximately two-thirds of the root ball height, the ties on the sacking shall be cut and the top portion of the burlap folded back carefully to avoid disturbing the integrity of the root ball. The sacking shall not be removed. The remainder of the hole will then be backfilled and firmed.

Where wire baskets are used to encase and support the root ball of supplied plant material, these shall not be removed. The top of the wire basket shall be cut away or completely folded back and buried without disturbing the integrity of the root ball.

Forestry Seedlings - Planting of forestry seedlings shall be in accordance with the Silvicultural Policy and Procedures Manual, Volume 3, of the British Columbia Ministry of Forests and Range.

Tree Support - All trees of a size requiring staking or guy wiring shall be supported in accordance with the details and instructions provided on SS Drawings SP754-04 through SP754-07. All hardware required shall be installed without damage to plants.

Trees that are dislodged during the Contract period shall be uprighted and re-secured as required. Trees that are damaged shall be replaced by the Contractor.

754.37 Seeding - Application of seed, fertilizer, and other materials shall be at the rates specified in the Special Provisions.

The requirements for construction shall be in accordance with DBSS 757 Revegetation Seeding.

754.38 Sodding - The required fertilizer shall be applied at the rates specified in the Contract and Design, and worked well into the topsoil prior to laying the sod.

Sod shall be laid within 24 hours after delivery unless proper storage arrangements can be made.

The sections of sod shall be laid close together with joints staggered. No open joints are to be visible, and no pieces are to overlap.

Sod shall be laid smooth and flush with the adjoining grass areas, adjacent hard surfacing, and the tops of curbs and planting bed liners, unless otherwise shown on the Design. All necessary cutting shall be done using sharp implements.

On slopes of approximately 2.5 to 1 and steeper, the sod shall be laid lengthwise across the slope, and the material secured with wooden stakes driven flush with the sod at intervals not exceeding 0.5 metres. There shall be at least three stakes per individual sod piece. On slopes of gradients between 2.5 and 1.5 to 1, the bottom three rows of sod and every third subsequent row shall be secured with stakes. For slopes steeper than 1.5 to 1, every sod course shall be staked.

Sodded areas shall be rolled or suitably tamped to ensure a good bond with the topsoil, and then subsequently protected from heavy foot traffic or equipment travel.

Unless otherwise indicated, sodded areas shall be evenly watered within 12 hours of installation, and with sufficient quantity to saturate the grass and the upper portion of the topsoil.

754.39 Watering - All trees, shrubs, groundcovers, vines, and designated grass areas, shall be watered immediately after planting, and regular watering shall continue as required for plant health until final acceptance of the work.

754.40 Pruning - Pruning shall be limited to the minimum necessary to remove dead or injured tissue and branches interfering with desirable growth habit and overall health of the plant.

Pruning shall be done in accordance with proper horticultural practice, using clean, sharp tools appropriate to the task and in a manner that preserves the natural character of the plant.

754.41 Mulching - When specified in the Design or the Contract, individual tree pits and planting beds shall be mulched.

Mulching of tree pits and planting beds shall be carried out after watering, to an even depth of 50 mm after settlement, unless otherwise specified on the Design or the Contract.

754.42 Clean-up - All plant containers and waste materials resulting from landscaping and planting operations shall be removed from the Site and appropriately disposed of.

754.43 Conditions for Acceptance - The Contractor shall ensure that the following conditions are met for all planted and grassed areas:

a) Topsoil quality, fertility levels, depths and surface conditions are as set out in the Design and Contract;

b) All plants are of the species and varieties specified and planted in the locations shown on Design;

c) All plants are healthy and growing vigorously. Seeded grass areas are sufficiently established into the underlying growing medium, are free of thin and bare patches, and are relatively free of weeds: not more than 5% in lawn areas, and not more than 15% in rough grass areas;

d) The water content in the topsoil is adequate, i.e. – when irrigation is provided,

e) Trees are supported, as specified;

f) Pruning is complete, in accordance with proper horticultural practice and to the satisfaction of the Designer and Quality Manager;

g) All planting beds and tree pits are free of weeds;

h) Mulch is in place, as required and;

i) Unmulched areas are cultivated to leave a loose, friable, water-permeable surface;

j) Maintenance procedures set out in DBSS 754.71 have been carried out.

MAINTENANCE

754.71 Maintenance - The following maintenance operations shall be performed as required ,from the time of landscape installation, until the Actual Completion Date :

a) Water shall be applied in sufficient quantity and by appropriate method to maintain optimum soil moisture conditions for healthy plant establishment, without causing surface soil erosion.

b) Weed control will be carried out, as required to prevent competition with establishing planted material and to maintain the aesthetic appearance of landscaped areas. The presence of weeds in plantation beds, individual planting pits, and designated lawn areas, is limited to a maximum of 5% of the surface area at any given time, unless otherwise stated in the Contract. The use of herbicides for the control of weed growth is not permitted.

c) Mowing shall be carried out at regular intervals, as required, to maintain grass in the areas designated, and at the height(s) specified in the Contract. Edges of areas designated “Lawn” shall be neatly trimmed. Excess clippings shall be removed immediately after mowing and trimming.

d) Fertility levels in planted and grassed areas shall be maintained in accordance with the requirements of the plant material.

e) Bark Mulch shall be maintained to the depth specified in the Design.

f) For non-mulched areas, the soil surface shall be cultivated, as required, to keep it loose and friable.

g) Insect and disease control shall be carried out as required. The Contractor shall secure all necessary Pesticide Use Permits in accordance with Ministry of Environment regulations, and possess a valid Pesticide Service License. When required, the Contractor shall provide a Pest Management Plan to MoE.

h) Establishment pruning to encourage proper shape and health of plants by removing dead, or broken and interfering branches and diseased or damaged tissue.

i) Maintenance of tree stakes, guy wires and tree ties to prevent plant dislodgement and damage to trunk and branches.

j) All plant material shall be alive and maintained in a healthy growing condition during the entire establishment period. Plant material which has died or is not healthy, and does not perform its function, shall be removed and replaced by the Contractor at the earliest opportunity, weather and season permitting. Grassed areas that show deterioration or bare spots shall be repaired immediately. All repair and/or replacement shall be in accordance with the original specifications and requirements.

SECTION 757

REVEGETATION SEEDING

DESCRIPTION

757.01 Scope - This Section refers to those portions of the work that are unique to the supply and application of seed, fertilizer, mulch, tackifier, and other materials used for revegetating disturbed areas, and that are not designated for treatment under DBSS 754, Planting of Trees, Shrubs, and Ground Covers. This Section must be referenced and interpreted simultaneously with all other Sections pertinent to the works described herein.

757.02 References - Guidelines for Hydroseeding in Proximity to Hydro Lines, Canada Seed Act, and BC Weed Control Act & Regulation.

MATERIALS

757.11 Handling and Storage - All seed, mulch, fertilizers and other dry materials shall be stored in a dry, weather proof storage place and shall be protected from damage by heat, moisture, rodents or other causes until the time of seeding. Supplier labels or other identification are not to be removed or defaced.

757.12 Seed

757.12.01 Supply of Seed - All seed specified shall be supplied by the Contractor and obtained from a recognized source.

757.12.02 Seed Type and Grade - All seed supplied either as individual species, or as a seed mix, shall comply with the requirements of the Canada Seed Act and Regulations, and the grade standards for that particular crop kind. Grass and legume seed shall meet or exceed Common No.1 grade prior to mixing with other species. Seed shall be free of propagules of plant species designated as noxious weeds under the BC Weed Control Act & Regulations.

All legume seed shall be inoculated with an adapted bacterial culture to ensure nitrogen fixation.

Seed mixes used for general roadside revegetation, and for the general conditions and areas indicated, shall be as shown on the table "Standard Grass Seed Mixes For Revegetation of British Columbia Highway Roadsides", unless otherwise specified in the Contract.

When specified, wildflower and shrub seed shall be supplied to the requirements of the Contract and Design.

757.12.03 Seed Analysis Report - The Contractor shall provide valid Certificates of Analysis for each species and seed lot used in a mix. These shall set out details of the seed as specified in the "Canadian Methods and Procedures for Testing Seed".

757.12.04 Packaging and Labelling - Seed shall be supplied in the original sealed packages, with legible labels securely attached, and providing the following information:

- Supplier's name and address
- Analysis of seed mixture – the grade, and the name and percentage by weight of individual seed species
- Percentage of Pure Live Seed (PLS) for each species
- Lot number and crop year for each species in the mix
- Net weight (mass)
- Date and location of packaging

757.13 Fertilizer - Fertilizer shall comply with the provisions of the Canada Fertilizers Act and Fertilizer Regulations. Fertilizer shall be supplied as noted on the table "Standard Grass Seed Mixes For Revegetation of British Columbia Highway Roadsides" unless otherwise specified in the Contract.

757.14 Hydraulic Mulch - Hydraulic mulch shall be a wood fibre type, specifically designed for hydraulic seeding, and having demonstrated satisfactory past performance for this purpose. The product shall be dyed green for appearance and ease of monitoring application.

Mulch shall be supplied in packages bearing the manufacturer's label, clearly indicating the weight and product name.

Mulch may contain a tackifier, which shall adhere to mulch to prevent separation during shipment and to avoid chemical agglomeration during mixing in hydraulic mulching equipment.

757.15 Water - Water used for hydraulic seeding operations shall be free of impurities that would inhibit germination and growth or may be harmful to the environment. Unless otherwise noted in the Contract, the Contractor shall be responsible for securing a water source for hydraulic application of materials, including obtaining use permits under the Water Act if water is to be drawn from waterbodies, and for all cost to supply.

757.16 Other Materials - Tackifiers, Bonded Fiber Matrix coverings, erosion control blankets, soil amendments and other materials shall be supplied to the specifications in the Contract.

EQUIPMENT

757.21 General - Equipment used shall be capable of applying the materials uniformly over the designated areas.

Equipment shall not cause soil rutting or other site damage.

757.22 Hydraulic Seeding/Mulching Equipment - Equipment shall have the tank volume identified by an identification plate or sticker, which shall be affixed in plain view.

The hydraulic seeder/mulcher shall be capable of sufficient agitation to mix the materials into a homogenous slurry, and to maintain the slurry in a homogeneous state until application.

Equipment shall be adequately sized to the task, to complete work efficiently within the time frame specified, and to permit application of materials without excess water being applied, or undue time lapse between operations. Hydraulic mulchers should be capable of producing slurry viscosities containing approximately 18 to 30kg of mulch per 500 litres of water.

Extension hoses or pipes shall be provided to reach areas not accessible from the hydraulic seeder.

CONSTRUCTION

757.31 Scheduling - Work shall be scheduled to ensure a minimum duration of on-Site storage of materials, minimum compaction of topsoil, and prompt mulching operations.

The work shall be co-ordinated with the schedule of other trades, and be well integrated with specific requirements such as Sediment and Drainage Management Plans, which may be provided for any given project.

757.32 Protection - Existing Site equipment, roadways, landscaping, reference points, monuments, markers, utilities and structures shall be protected from damage.

757.33 Timing of Material Application - Material application shall be carried out in accordance with the milestone dates provided in the Contract, and after fine grading has been completed and the areas prepared.

757.34 Methods - The methods chosen for material application shall be at the Contractor's discretion, unless otherwise specified in the Contract.

757.35 Rates of Application - Application of fertilizers, seed mixtures, mulch and other materials shall be at the rates specified in the Contract and by the Designer.

757.36 Record of Application - The Contractor shall maintain a record of all pertinent application information on the form supplied by the Ministry for this purpose, or similarly provided by the Contractor. Refer to Sample Form "Daily Seeding/Application Record".

757.37 Application Method for Mechanical Drop or Broadcast Dry Seeding - Seed shall be applied in two intersecting directions, except where conditions dictate seeding in one direction only.

Seeding shall overlap adjoining ground cover by 300mm.

Refer to the Contract for specific instructions for installation of wildflower, shrub and other seed as may be applicable.

757.38 Hydraulic Application of Materials

757.38.01 General - The hydraulic seeder/mulcher shall be operated in compliance with Ministry safety standards including those detailed in the publication "Guidelines for Hydroseeding in Proximity to Hydro Lines."

Materials shall not be sprayed on objects not expected to support plant growth.

The Contractor shall be responsible for any overspray or damages incurred during hydroseeding.

757.38.02 Mixing - The required quantities of seed, fertilizer, mulch, tackifier and other material shall be charged into the tank accurately by weight or by an acceptable system of mass calibrated volume measurement.

The materials shall be thoroughly mixed into a homogeneous water slurry prior to application.

All seed shall be added last when mixing. Pellet inoculated seed shall be applied immediately after placement into tank, and if this is not possible, dry application methods must be used. Other seed shall not be left in the tank for unreasonable lengths of time prior to application, i.e. – exceeding one or two hours, particularly when in contact with fertilizer solution.

757.38.03 Application - The mulch and tackifier components of hydraulically applied mixtures will generally be applied in stages. The initial pass of the hydraulic seeder will distribute the correct amount of seed and fertilizer for the area being done, as well as up to one third of the required mulch/tackifier. The subsequent pass(es) will complete the mulching/tacking process to the required rate.

Mulch shall be applied to form an even, uniform mat blended 150 mm into adjacent vegetated areas or previous mulch applications.

757.39 Related Work - Additional related work such as the application of erosion control blanket or other coverings, and harrowing or discing of soil following material application, shall be as specified in the Contract and by the Design.

757.40 Clean-up - All surplus and waste materials resulting from seeding operations shall be removed from the job site..

Hydraulic seeding and/or mulching overspray that may cause problems on areas or objects not designated for re-vegetation, shall be removed in an appropriate manner.

757.41 Not Used

757.42 Repairs - Seeded areas that show thin application or bare spots shall be re-treated with the specified materials at the Contractor's earliest opportunity, weather and season permitting.

SHEET #_____

MINISTRY OF TRANSPORTATION AND HIGHWAYS

PROJECT #_____

DAILY SEEDING/APPLICATION RECORD

DATE	LOAD #	AREA COVERED IN HECTARES				SEED APPLIED* (IN KG.)			OTHER MATERIALS* (IN KG.)				REMARKS
		GRASS	MULCH	REFERT		GRASS	RYE	OTHER	FERT	MULCH	TACK	MISC.	
	1												
	2												
	3												
	4												
	5												
	6												
	7												
	8												
	9												
	10												
	11												
	12												
	13												
	14												
	15												
	16												
	17												
	18												
	19												
	20												
Total of Sheet													

Provide details of seed mixes and other materials, as required, in this space:

THIS SHEET CERTIFIED CORRECT:

FOR MINISTRY _____

FOR CONTRACTOR _____

DATE :